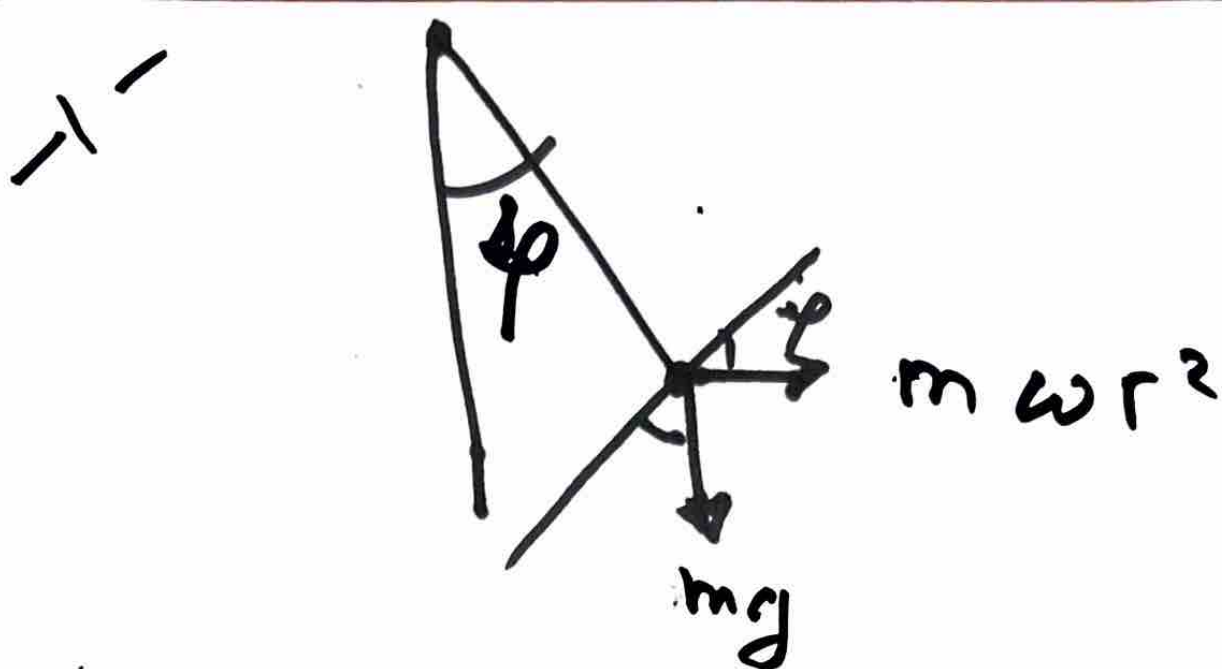
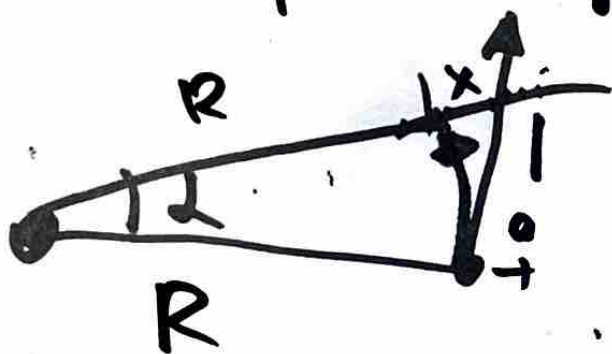




$$a_{\text{cen}} = \frac{v^2}{R} = \omega^2 R$$

centripetal.



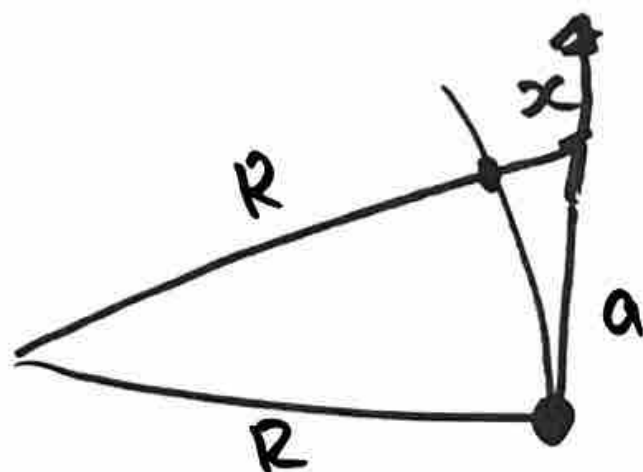
$$d = \omega dt$$

$$v = \omega R$$

$$a = v dt = \omega R dt$$

$$x = \sqrt{R^2 + a^2} - R$$

-2-



$$x = \sqrt{R^2 + a^2} - R =$$

$$= \sqrt{R^2 + (v dt)^2} - R$$

$$= \sqrt{R^2 + (\omega R dt)^2} - R$$

$$= \sqrt{R^2 + \omega^2 R^2 (dt)^2} - R =$$

$$= R \sqrt{1 + \omega^2 (dt)^2} - R$$

$$\boxed{\sqrt{1+x} \approx 1 + \frac{x}{2}}$$

$$|x| < 1$$

$$a = R \left(1 + \frac{\omega^2 (dt)^2}{2} \right) - R$$

$$= \frac{R \omega^2 (dt)^2}{2}$$

$$f(x) = \sqrt{x} \quad \boxed{-3-}$$

$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$\begin{aligned} f(1+x) &= f(1) + f'(1) \cdot x \\ &= 1 + \frac{1}{2} x \end{aligned}$$

$$a = \frac{R (\omega)^2 (dt)^2}{Z}$$

$$\text{Distance} = \text{Acceleration} \cdot \frac{t^2}{2}$$

$$x = x_0 + v_0 t + \frac{a t^2}{2}$$

$$L = L + \frac{L}{T} \cdot T + \frac{L}{T^2} \cdot T^2$$

$$\text{Acceleration} = \omega^2 R = \frac{L}{T^2}$$

-5-

$$m R \ddot{\varphi} + b \dot{\varphi} = -mg \sin \varphi + m \omega^2 R \sin \varphi \cos \varphi$$

$$\ddot{\varphi} \rightarrow 0$$

$$b \dot{\varphi} = mg \sin \varphi \left[-1 + \frac{\omega^2 R}{g} \cos \varphi \right]$$

Fixed points

$$\varphi = \pi n, n = 0, \pm 1, \pm 2, \pm 3, \dots$$

$$\cos \varphi = g / \omega^2 R$$

$$\varphi = \pm \text{ArcCos} \left[\frac{g}{\omega^2 R} \right]$$

$$\frac{\omega^2 R}{g} \equiv \gamma \quad [\gamma] = 1$$

$$b \dot{\varphi} = mg \sin \varphi (-1 + \gamma \cos \varphi)$$

-b-

$$b\ddot{\varphi} = (mg \sin \varphi) [-1 + \gamma \cos \varphi]$$

$$b\ddot{\varphi} = f(\varphi, \gamma) \quad \boxed{\gamma \cos \varphi = 1}$$

$$\frac{\partial f}{\partial \varphi} = mg \cos \varphi (-1 + \gamma \cos \varphi) \\ \neq mg \sin^2 \varphi \cdot \gamma$$

$$= -mg \cos \varphi + mg \gamma \cos^2 \varphi \\ - mg \gamma \sin^2 \varphi$$

$$\sin^2 x + \cos^2 x = 1$$

$$\sin^2 \gamma = 1 - \cos^2 x$$

$$\frac{\partial f}{\partial \varphi} = -mg \cos \varphi + mg \gamma \cos^2 \varphi \\ - mg \gamma (1 - \cos^2 \varphi)$$

7-

$$\begin{aligned}\frac{\partial f}{\partial \varphi} &= -mg \cos \varphi + mg \gamma \cos^2 \varphi \\ &\quad - mg \gamma + mg \gamma \cos^2 \varphi \\ &= -mg \cos \varphi - mg \gamma + 2mg \gamma \cos^2 \varphi\end{aligned}$$

$\varphi = 0$:

$$\begin{aligned}\left. \frac{\partial f}{\partial \varphi} \right|_{\varphi=0} &= -mg - mg \gamma + 2mg \gamma \\ &= mg[-1 + \gamma]\end{aligned}$$

$0 < \gamma < 1 \rightarrow \text{stable}$

$\gamma > 1$, unstable

$$\gamma = \frac{\omega^2 R}{g}$$

$\gamma \cos \varphi = 1$, as $\varphi = 1/\gamma$

• $\varphi = \pm \text{Arc Cos } 1/\gamma$

$$\left. \frac{\partial f}{\partial \varphi} \right|_{\gamma \cos \varphi = 1} = -mg \frac{1}{\gamma} - mg \gamma + 2mg \frac{\gamma}{\gamma}$$

- \delta -

$$\left. \frac{\partial f}{\partial \varphi} \right|_{\gamma \cos \varphi = 1} = -\frac{mg}{\gamma} - mg\gamma + \frac{2mg}{\gamma}$$

$$= mg\left(\frac{1}{\gamma} - \gamma\right) = \frac{mg}{\gamma}(1 - \gamma^2)$$

if $\gamma > 1$ - stable

if $\gamma < 1$, - unstable

-9-

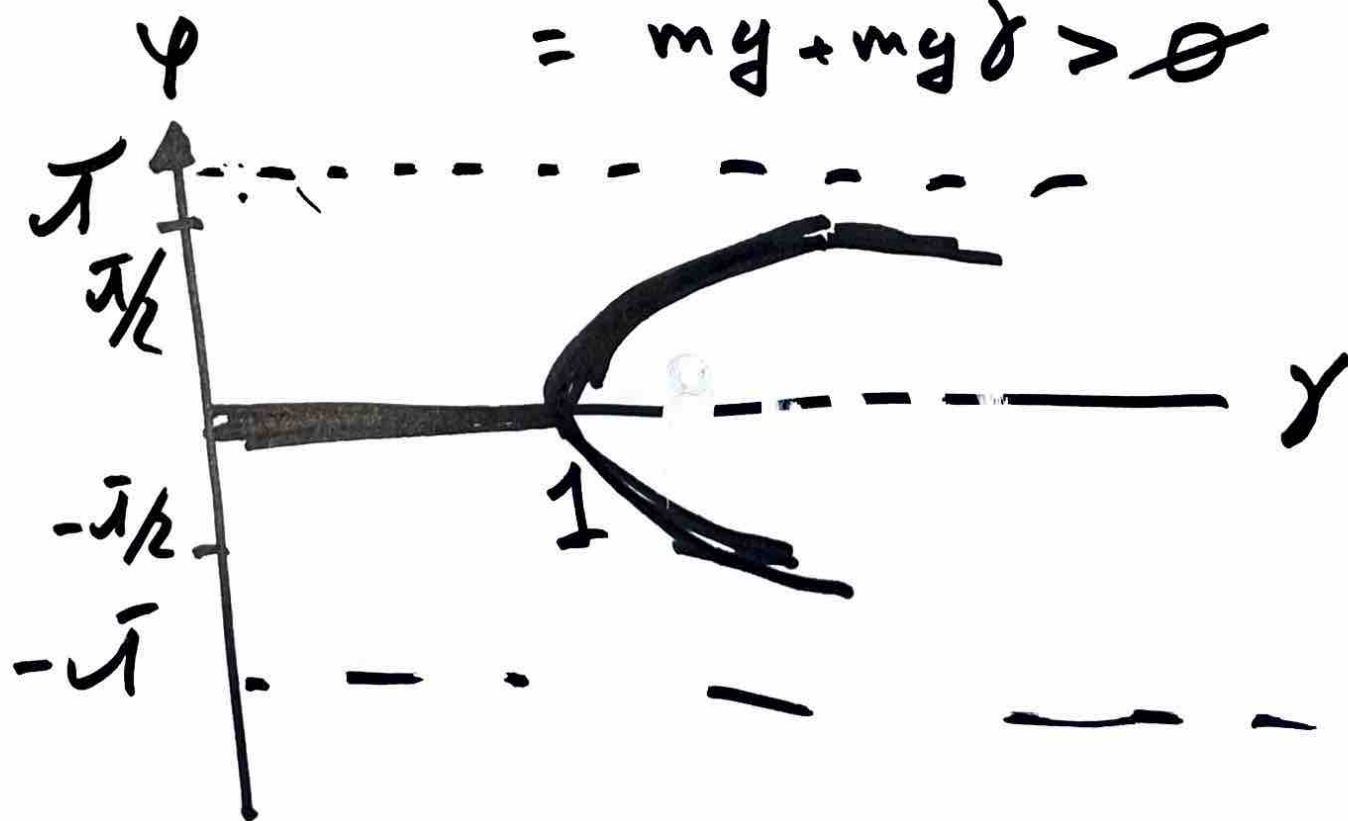
$$\varphi = \pi$$

$$\frac{\partial f}{\partial \varphi} = -mg \cos \varphi - mg\delta + 2mg\delta \cos^2 \varphi$$

$$\cos \pi = -1$$

$$\left. \frac{\partial f}{\partial \varphi} \right|_{\varphi=\pi} = mg - mg\delta + 2mg\delta$$

$$= mg + mg\delta > 0$$



$$m R \ddot{\varphi} + b \dot{\varphi} = -mg \sin \varphi + m R \omega^2 \sin \varphi \cos \varphi$$

$$\text{Newtons} = \text{kg} \cdot \frac{\text{Meter}}{\text{Sec}^2}$$

$$\text{Force} = \text{Mass} \cdot \frac{\text{Length}}{\text{Time}^2}$$

$$[m R \ddot{\varphi}] = \text{Mass} \cdot \frac{\text{Length}}{\text{Time}^2}$$

$$[-mg \sin \varphi] = \text{Mass} \cdot \frac{\text{Length}}{\text{Time}^2}$$

$$[m R \omega^2] = \text{mass} \cdot \frac{\text{Length}}{\text{Time}^2}$$

$$[b] = \frac{\text{mass} \cdot \text{Length}}{\text{Time}}$$

$$L = T \cdot r ; [L] = T \text{ me}$$

$$[r] = 1$$

$$[T] = T \text{ me}$$

$$\frac{d}{dt} = \frac{dr}{dt} \frac{d}{dr} = \frac{1}{r} \frac{d}{dr} ;$$

$$\frac{m R}{T^2} \frac{d^2}{dr^2} \varphi + \frac{6}{T} \frac{d}{dr} \varphi = -mg \sin \varphi + m R \omega^2 \sin \varphi \cos \varphi$$

$$\frac{R}{T^2} g \frac{d^2}{dr^2} \varphi + \frac{6}{T} mg \frac{d}{dr} \varphi = -\sin \varphi + \frac{R \omega^2}{g} \sin \varphi \cos \varphi$$

$$\frac{g}{Tmg} = 1, T = \frac{g}{mg}$$

$$\frac{R m^2 g}{g^2} \frac{d^2}{dt^2} \varphi(r) + \frac{d}{dt} \varphi(r) = -\sin \varphi + \frac{R \omega^2}{g} \sin \varphi \cos \varphi$$

$$\mathcal{E} = \frac{R m^2 g}{g^2} \quad \gamma = \frac{R \omega^2}{g}$$

$$\mathcal{E} \ddot{\varphi} + \dot{\varphi} = -\sin \varphi + \gamma \sin \varphi \cos \varphi$$

$$[\mathcal{E}] = \frac{\text{Length Mass}^2 \text{Length Time}^2}{\text{Time}^2 \text{mass}^2 \text{Length}^2}$$

13

$$\text{if } \varepsilon < 1, \quad \varepsilon \frac{d^2 \varphi}{dt^2} < \frac{d\varphi}{dt}$$

$$\begin{aligned} \ddot{\varphi} &= -\sin \varphi + \gamma \sin \varphi \cos \varphi; \quad \varphi(t=t_0) = \varphi_0 \\ \varepsilon \ddot{\varphi} + \dot{\varphi} &= -\sin \varphi + \gamma \sin \varphi \cos \varphi \quad \varphi(t=t_0) = \varphi_0 \\ \dot{\varphi} &= \varphi_0 \quad \varphi(t=t_0) = \varphi_0 \end{aligned}$$

$$\varepsilon \dot{\varphi} + \varphi = -\sin \varphi + \gamma \sin \varphi \cos \varphi$$

$$\dot{\varphi} = \frac{1}{\varepsilon} (-\varphi - \sin \varphi + \gamma \sin \varphi \cos \varphi)$$

$$\dot{\varphi} = \varphi$$

$$\varphi = -\sin \varphi + \gamma \sin \varphi \cos \varphi$$

The limit $\epsilon \rightarrow 0$ corresponds to strongly overdamped system, and will quickly revert to first order behavior.

Moreover, transient will be shorter for smaller ϵ or ϵ smaller.

-15

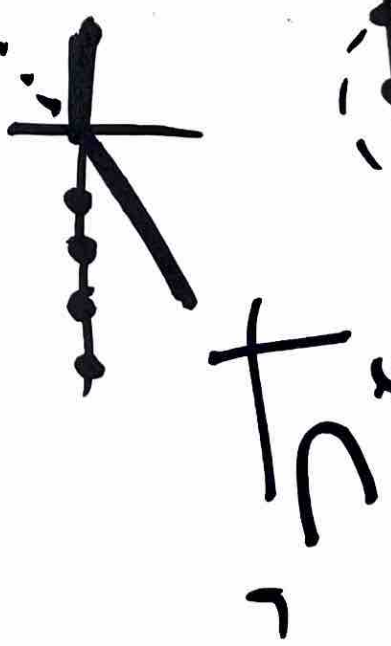
Saddle node

$$\dot{x} = r + x^2$$



Transcritical

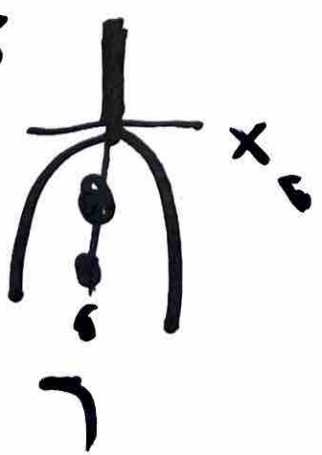
$$\dot{x} = x(r - x)$$



Pitchfork Bifurcation

$$\dot{x} = x(r - x^2)$$

Supercritical



$$\dot{x} = x(x^2 + r)$$

Subcritical



$$\dot{x} = x(r + x^3)$$



-16-

Chapter 4

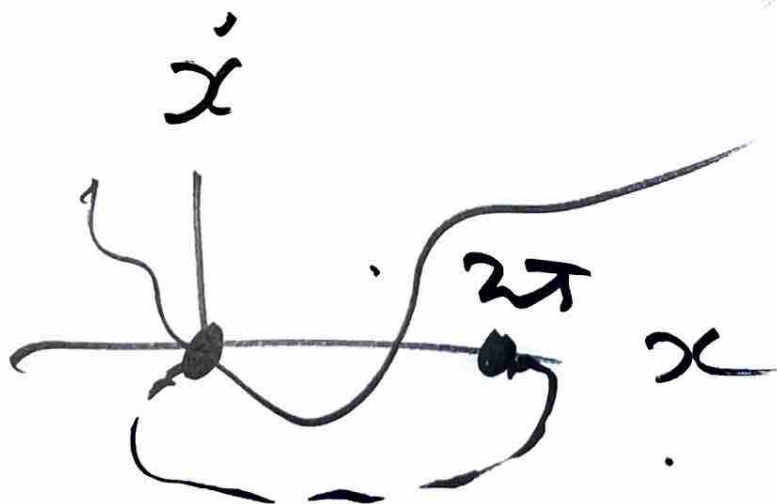
Flow on a circle

$$\dot{x} = f(x) \quad \text{Ch 2}$$

$$\dot{x} = f(x, r) \quad \begin{array}{l} \text{tunable} \\ \text{parameter} \end{array}$$

$$\dot{\theta} = f(\theta) \quad \theta$$

$$\theta \equiv \text{mod} (0, 2\pi)$$



-12-

$$\ddot{\theta} = f(\theta)$$

$$f(\theta + 2\pi) = f(\theta)$$

$f(\theta)$ is two π periodic

$$m R \ddot{\theta} + b \dot{\theta} + m g \sin \theta = 0$$

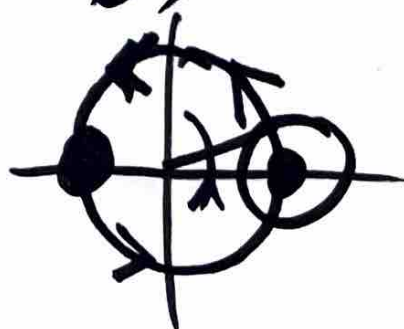
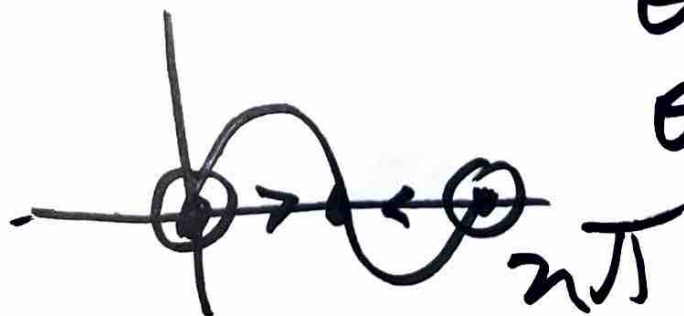
$$\ddot{x} = \sin x$$

Limit to $\theta = \text{mod}(x, 2\pi)$

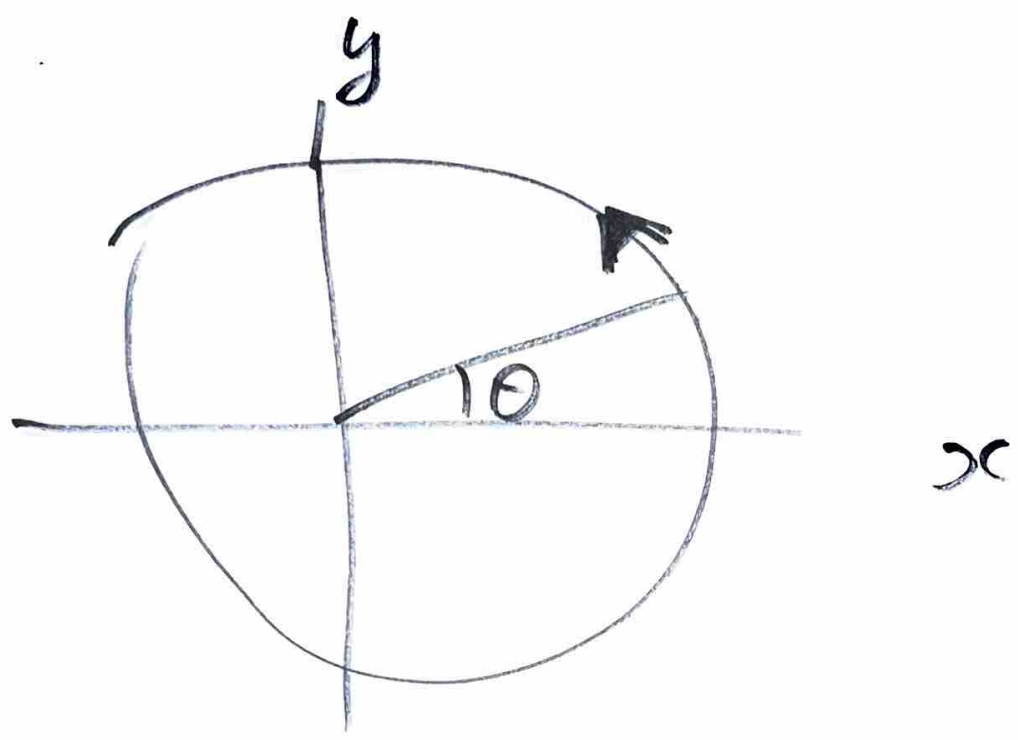
$$\ddot{\theta} = \sin \theta$$

$\theta = \pi$ stable

$\theta = 0$, unstable



-18-



-19-

$$\dot{\theta}(t) = \omega$$

$$\theta(t) = \omega t + \theta_0$$

20- Example

Two RUNNERS are

running on a track,

Runner 1 can do it in

T_1 seconds

Runner 2 can do track

in T_2

seconds

IF $\theta_1(t=0) = \theta_2(t=0) = 0$
when do they meet
again?

IF $T_1 < T_2$ when Runner 1
overtakes runner 2

-21-

$$\dot{\theta}_1 = \frac{2\pi}{T_1}, \quad T_1 - \text{lap time}$$

$$\dot{\theta}_2 = \frac{2\pi}{T_2}$$

what is T such

$$\text{that } \theta_1(T) = \theta_2(T)$$

$$\text{if } \theta_1(t=0) = \theta_2(t=0) = 0$$

$$\dot{\theta}_1 = \frac{2\pi}{T_1} \Rightarrow \theta_1 = \frac{2\pi}{T_1} t + \theta_1(t=0)$$

$$\theta_2 = \frac{2\pi}{T_2} t + \theta_2(t=0)$$

$$\theta_1(T) = \theta_2(T) = 2\pi$$

$$T \left(\frac{2\pi}{T_1} - \frac{2\pi}{T_2} \right) = 2\pi$$

$$T = \frac{1}{1/T_1 - 1/T_2} = \frac{T_1 T_2}{T_2 - T_1}$$